ALIMA Simply — Installer manual

CB ALIMA · Version October 2026

CB ALIMA · Electronic design for adapted (PMR) vehicles · cb-alima.fr · contact@cb-alima.fr

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English translation provided for information. In case of any difference, only the French version of this manual is binding.

Overview

The ALIMA Simply is a button-operated auxiliary control board: it drives up to 6 vehicle functions (relays) from 4 switches, for drivers who cannot use the original controls.

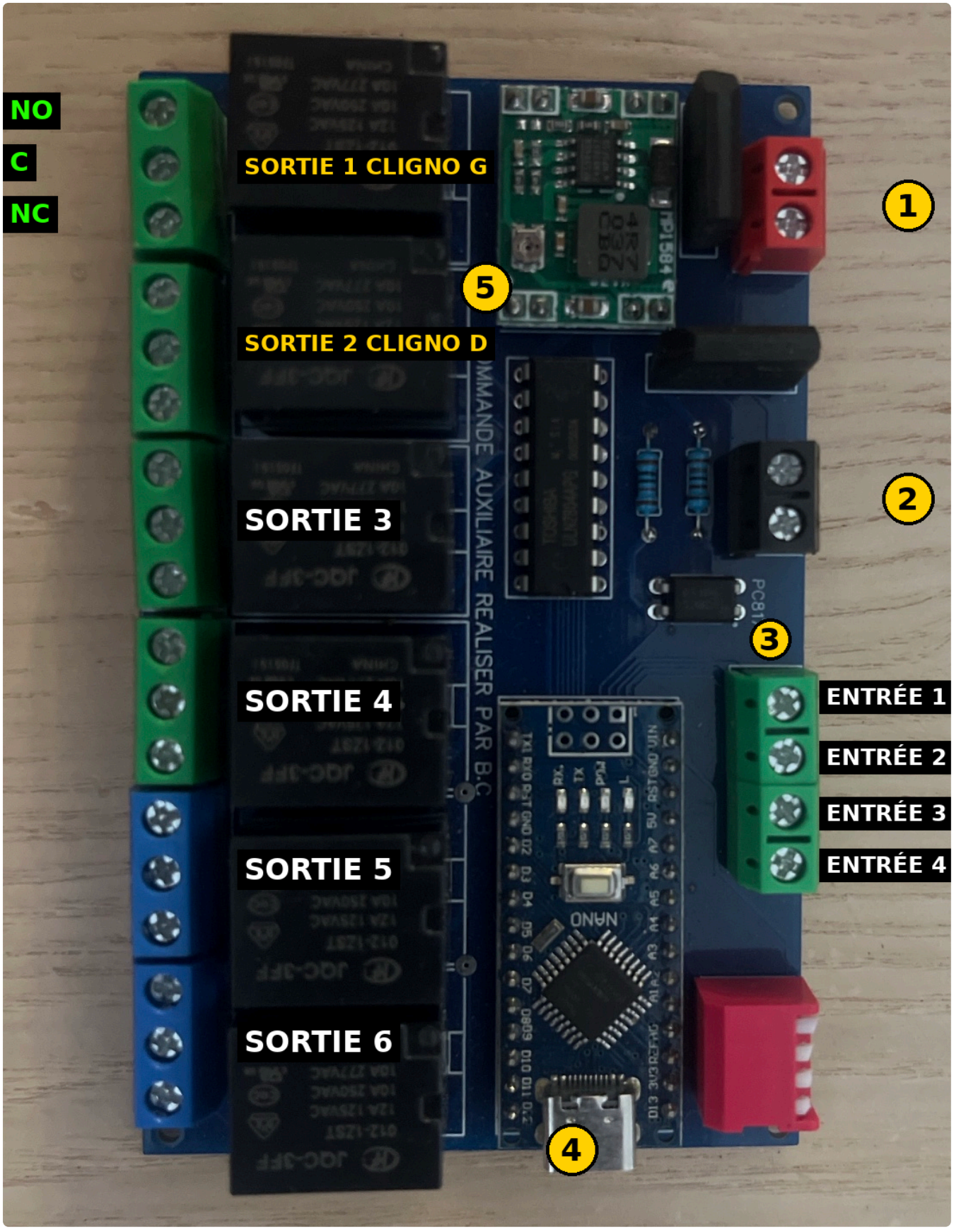
The names ENTRÉE (input) and SORTIE (output) are kept in French in this manual, exactly as they are printed on the plastic housing and shown in the ALIMA application.

Item	Specification
Microcontroller	Arduino Nano (ATmega328P)
Power supply	Vehicle 12 V, no polarity (diode bridge)

Item	Specification
Switch inputs	ENTRÉE 1 to ENTRÉE 4, connected to ground by the switch
Lights detection input	ENTRÉE 12 V (feux), no polarity, on the black terminal block
Outputs	SORTIE 1 to SORTIE 6 (relays). SORTIE 1 and 2: left and right turn signals (and hazard lights). SORTIE 3 to 6: free
Configuration link	USB-C (serial)

Board layout

Board seen from above, as in the photo (USB-C at the bottom). The names ENTRÉE 1…, SORTIE 1… and the markings C / NO / NC are those used in the ALIMA application: write them identically on the plastic housing. Terminal block colours are standardised.


NO
C
NC
SORTIE 1 CLIGNO G
SORTIE 2 CLIGNO D
SORTIE 3
SORTIE 4
SORTIE 5
SORTIE 6
1
2
3
4
5
ENTRÉE 1
ENTRÉE 2
ENTRÉE 3
ENTRÉE 4
IMMANDE AUXILIAIRE REALISER PAR B.C

Mark	Terminal block	Name in the app	Role
1	Red, 2 screws	—	12 V power supply. Diode bridge: direction does not matter.
2	Black, 2 screws	ENTRÉE 12 V (feux)	Lights detection, on the two black pins, in parallel with a light (dipped beam, side lights, or number plate light if the front lights are multiplexed). No polarity.
3	Green, 4 screws	ENTRÉE 1 to ENTRÉE 4 (ENTRÉE 1 at the top in the photo)	Switches. Each input is connected to ground through the switch contact.
4	USB-C socket	—	Configuration with the ALIMA application.
5	Voltage regulator	—	Built-in, sealed potentiometer: do not touch.
—	6 terminal blocks, 3 screws (top to bottom: 4 green, then 2 blue)	SORTIE 1 to SORTIE 6 (SORTIE 1 at the top in the photo)	Relay outputs. Screw order, top to bottom in the photo: NO, C (common), NC. SORTIE 1 = left turn signal, SORTIE 2 = right turn signal: these two functions are fixed. SORTIE 3 to 6 are free.

Safety warnings

Installation must be carried out by a vehicle adaptation professional. A wiring error can cause a fire, the malfunction of a safety function or loss of control of the vehicle.

- Disconnect the negative battery terminal before any work, and wait for the circuits to discharge (airbags in particular).
- Never connect the board to a circuit related to active or passive safety (braking, steering, airbags, ABS/ESP) without ALIMA's written approval.
- Never directly control a light, a wiper or a regulated function without checking that the adaptation is allowed for this vehicle.

- The vehicle's original control must remain usable and have priority: the board must never prevent it from working.
- Each relay output must be protected by its own fuse, placed as close as possible to the power source.
- Do not exceed the maximum current of the relay contacts (10 A). For a higher load, use an external power relay.
- Never connect the USB cable to the computer while the vehicle is running or the board is driving loads: configure at standstill.
- Mount the board protected from water, heat and vibration, in the supplied housing, away from ignition cables and motors.
- Route cables in sleeves, with grommets where they pass through sheet metal, away from moving and hot parts.
- No modification of the board, its components or its firmware is accepted.
- It is forbidden to touch the built-in, sealed potentiometer on the voltage regulator.
- Give the customer the user instructions and explain how the installation works and its limits.
- Connections must be soldered or crimped, then insulated with heat-shrink sleeving. No self-stripping connectors ("Scotchlok" type), no tapping into airbag harnesses (yellow sleeving) or vehicle communication lines (CAN, LIN, multiplexing).
- After installation, test every function with the vehicle stationary and the engine running, then during a road test carried out by the installer before handing over the vehicle.

Fuses and cable sizes

The fuse rating is chosen to protect the cable, not the load: it must be higher than the load current and lower than what the cable and the relay contact can withstand.

Circuit	Fuse	Cable size
Board power supply (red terminal block)	2 A	0.75 mm²
ENTRÉE 1 to ENTRÉE 4 (green terminal block)	No dedicated fuse	0.75 mm²
ENTRÉE 12 V (feux) (black terminal block)	Fuse of the original lighting circuit	0.75 mm²

Circuit	Fuse	Cable size
SORTIE 1 to SORTIE 6 (turn signals included)	Rated by the installer according to the load (table below), 10 A maximum	According to the load and the fuse (table below)

Each relay contact withstands 10 A maximum (value printed on the relay). Never fit a fuse above 10 A on an output, and never above 5 A if the cable stays at 0.75 mm².

Choosing the fuse for each output (12 V system). Current = power ÷ 12. The fuse must stay above the load current (at least 20 % margin) and never exceed what the cable can withstand.

Load on the output	Current	Minimum cable size	Recommended fuse
up to 24 W	2 A	0.75 mm²	3 A
up to 42 W	3.5 A	0.75 mm²	5 A
up to 72 W	6 A	1 mm²	7.5 A
up to 96 W	8 A	1 mm² (1.5 mm² recommended)	10 A
more than 96 W	more than 8 A	not allowed on the board	use an external power relay driven by the output

These values are indicative for short copper cables (less than 3 m). For a load with a high inrush current (motor, pump), use the starting current, or an external relay.

Indicative guide for automotive copper cable, to be confirmed according to the cable and the length of the run:

Cable size	Maximum fuse
0.75 mm²	5 A
1 mm²	7.5 A to 10 A
1.5 mm²	15 A
2.5 mm²	20 A
4 mm²	30 A

Fuse holder less than 30 cm from the source (battery or fuse box). Never replace a fuse with a higher rating.

Connecting the board

Battery disconnected during all wiring. Reconnect only once everything has been checked.

1. Fix the board in its plastic housing, protected from water and heat.
2. Power supply (mark 1, red terminal block): connect 12 V and ground with a 2 A fuse, in 0.75 mm². Direction does not matter.
3. ENTRÉE 12 V (feux) (mark 2, black terminal block): connect the two black pins in parallel with a light (dipped beam, side lights, or number plate light if the front lights are multiplexed). Direction does not matter.
4. ENTRÉE 1 to ENTRÉE 4 (mark 3, green terminal block): each screw is one input. The switch contact connects the screw to ground.
5. SORTIE 1 (left turn signal) and SORTIE 2 (right turn signal): connect to C and NO (or NC depending on the installation), with their fuse. These two outputs are reserved for the turn signals and hazard lights.
6. SORTIE 3 to SORTIE 6: connect each load to C and NO (or NC as required), with its fuse. Screw order: NO, C, NC.
7. Reconnect the battery, then follow the commissioning procedure with the ALIMA application (next section) to test each output.

Terminal block	Colour	Screws	Connection
Power supply	Red	2	12 V and ground, 2 A fuse, no polarity
ENTRÉE 12 V (feux)	Black	2	In parallel with a light, no polarity
ENTRÉE 1 to ENTRÉE 4	Green	4	To ground through the switch
SORTIE 1 (left turn signal)	Green	3	NO / C / NC
SORTIE 2 (right turn signal)	Green	3	NO / C / NC
SORTIE 3 and SORTIE 4	Green	3 each	NO / C / NC
SORTIE 5 and SORTIE 6	Blue	3 each	NO / C / NC

Configuration

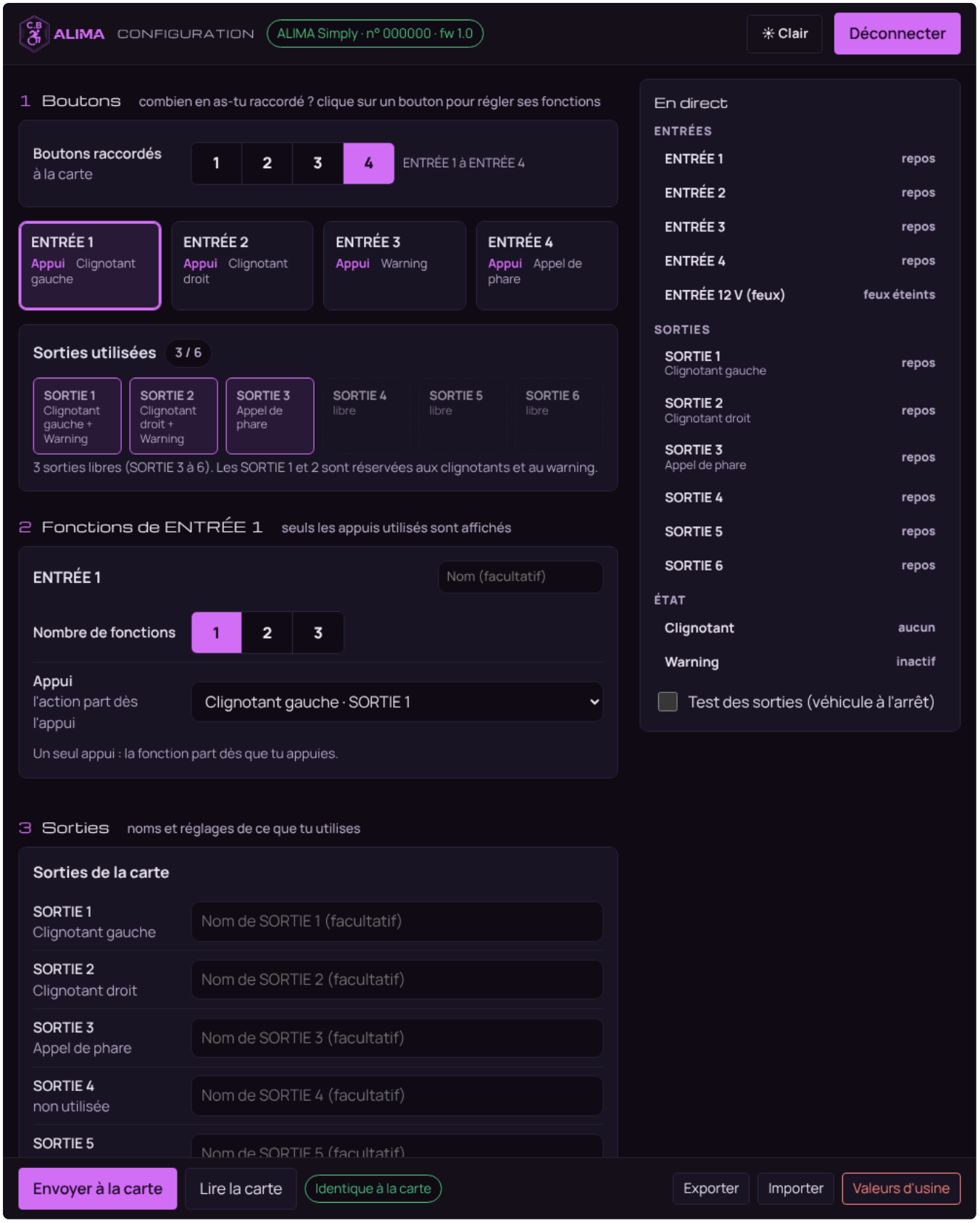
The board is configured through the USB-C socket (mark 4) with the ALIMA application, vehicle stationary. The application uses the same names as the housing: ENTRÉE 1 to 4, ENTRÉE 12 V (feux), SORTIE 1 to 6. The application screens are shown in French.

Commissioning with the ALIMA application

Step 1. Connect the board. Power the board, then connect it to the computer with a USB-C data cable (not a charge-only cable). Open the ALIMA application (Chrome or Edge, or the desktop app) and press “Connecter” (Connect). Opening the port restarts the board: the outputs drop out. Never connect while driving.



Step 2. Check recognition. The board is recognised automatically. The green badge shows the model (ALIMA Simply), the serial number and the firmware version. The application reads the configuration stored in the board. On the right, the “En direct” (Live) panel shows the state of each ENTRÉE, of ENTRÉE 12 V (feux) and of each SORTIE.



Factory configuration: ENTRÉE 1 = left turn signal, ENTRÉE 2 = right turn signal, ENTRÉE 3 = hazard lights, ENTRÉE 4 = headlight flash.

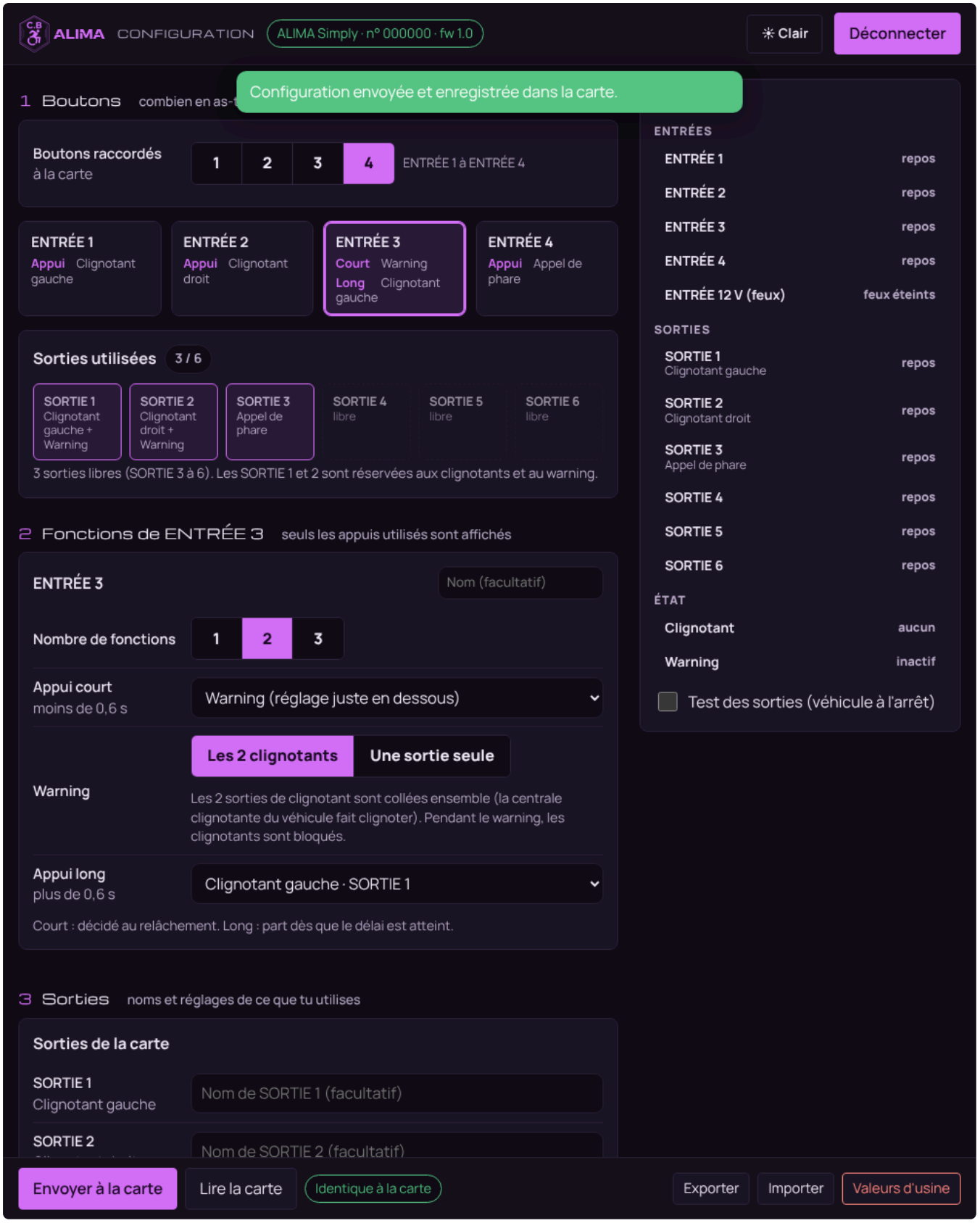
Step 3. Set the buttons (sections 1 and 2). Choose the number of connected buttons (1 to 4): they match ENTRÉE 1 to ENTRÉE n. Click the tile of an ENTRÉE, choose its number of functions (1 to 3: short, long, very long press) and the function of each press from the list.

Each function shows the SORTIE it controls. The “Sorties utilisées” (Outputs used) panel shows how many of the 6 SORTIES are taken; SORTIE 1 and 2 are reserved for the turn signals and hazard lights.

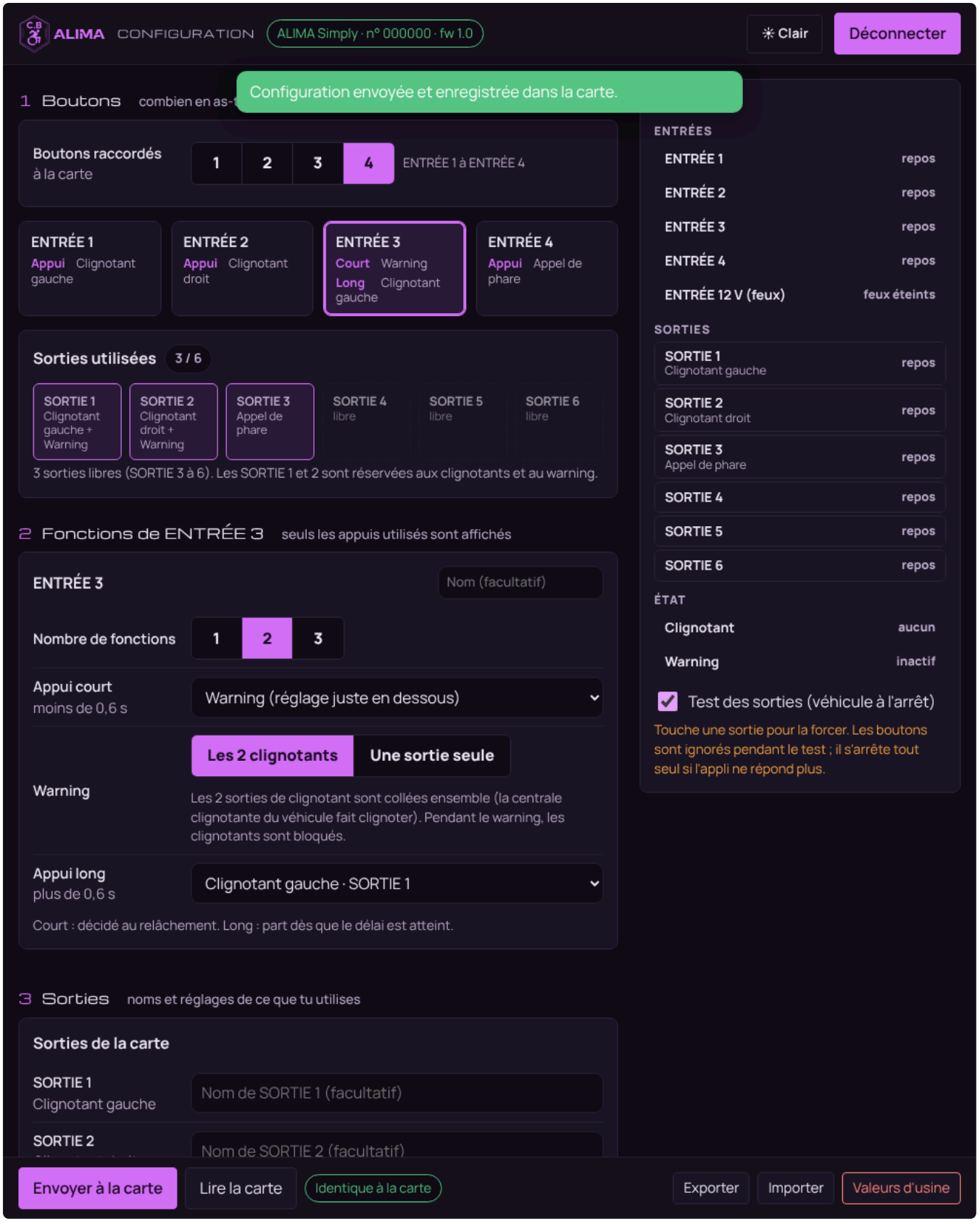
- **Hazard lights**: either the 2 turn signal SORTIES energised together, or a dedicated SORTIE. While the hazard lights are on, the turn signals are locked.
- **Headlight flash / main beam**: lights off = headlight flash, lights on = main beam. The lights are read on ENTRÉE 12 V (feux).
- **SORTIE 3 to 6**: latching (stays on), momentary (held while pressed) or pulse.

Step 4. Names and settings (sections 3 and 4). ENTRÉE and SORTIE names are optional and stay on the device. Settings: long press from 600 ms (minimum 200 ms), very long press from 2000 ms, default pulse 250 ms (50 to 2500 ms), and a box to invert ENTRÉE 12 V (feux).

Step 5. Send to the board. Press “Envoyer à la carte” (Send to board). The message “Configuration envoyée et enregistrée dans la carte” (configuration sent and saved in the board) appears and the badge changes to “Identique à la carte” (same as board). If a red “À corriger avant d'envoyer” (to fix before sending) frame appears, correct the point shown: sending is blocked while an error remains.



Step 6. Test the outputs, vehicle stationary. Tick “Test des sorties” (Output test), then touch a SORTIE to force it on and check that the right load responds. The buttons are ignored during the test, which stops by itself if the application stops responding.



Step 7. Check. Untick the test, unplug the USB cable, then try each ENTRÉE with its switch and check the functions one by one.

The “Exporter” (Export) and “Importer” (Import) buttons save the configuration to a file and copy it to another board. “Valeurs d'usine” (Factory settings) restores the original configuration. If the board is locked by ALIMA, the application switches to read-only. The

firmware is installed by ALIMA: do not modify it.

Checks before handing over the vehicle

- [ ] Wiring visually checked: no pinched or stripped cable, no contact with hot or moving parts.
- [ ] All fuses fitted with the correct rating.
- [ ] Each ENTRÉE (1 to 4) triggers the correct function with its switch.
- [ ] Each SORTIE controls the correct load, and nothing else (application output test).
- [ ] The vehicle's original control still works.
- [ ] ENTRÉE 12 V (feux): the function responds correctly with the lights on, then off.
- [ ] No warning light on the dashboard, no abnormal battery drain when parked.
- [ ] Manual given to the customer and operation explained.

Symptom	Check
Board does not respond	Power fuse, ground, +12 V supply
A relay does not engage	Output fuse, load wiring
Function triggers by itself	Interference: move cables away, check the ground

Installation record

To be filled in and signed by the installer for each vehicle, then kept with the configuration file exported from the ALIMA application. It covers the identification of the vehicle and the board (VIN, serial number, firmware version), the function and fuse of each SORTIE, the checks before handover, the road test and the information given to the customer.

The record is the **last page of this manual**, alone on its page so it can be printed separately. Its fields can be filled in directly in a PDF reader (Adobe Acrobat Reader, web browser). It can also be downloaded on its own: ALIMA Simply installation record (fillable PDF).

Send it to contact@cb-alima.fr within 10 days of installation, together with the exported configuration file: this is the condition for the 2-year warranty (see “2-year warranty”).

Liability and limits of use

Installation must be carried out by a qualified professional. By installing the board, the installer acknowledges having read this manual and accepts its conditions.

- The installer is solely responsible for the installation: choice and fitting of fuses, cable sizes, quality of connections, tests and handover of the vehicle to the customer.
- The installer is responsible for compliance with the regulations applicable to the modified vehicle (inspection, declaration or type approval of the adaptation, informing the customer's insurer, where applicable).
- The ALIMA Simply board is not a safety device. It must never be connected to a safety circuit (braking, steering, airbag, traction). Tests are carried out with the vehicle stationary.
- Any modification of the board, its components or its firmware is prohibited, including adjusting the sealed potentiometer of the voltage regulator. It voids the warranty and releases CB ALIMA from all liability.
- CB ALIMA cannot be held liable for damage resulting from an installation that does not comply with this manual, misuse, overloading of the outputs or modification of the board.
- Within the limits permitted by law, CB ALIMA is not liable for indirect damage (vehicle downtime, loss of business, removal and refitting costs, etc.). Its liability is limited, at its option, to the repair, replacement or refund of the board.
- The ALIMA Simply board is intended exclusively for installation professionals. It is not sold to private individuals for self-installation.
- The board configuration (choice of functions, assignment of ENTRÉES and SORTIES, timings) is carried out and validated by the installer, who is responsible for it. The ALIMA application is provided as is, as a configuration tool.
- The installer fills in the installation record, keeps it with the exported configuration file, and provides it to CB ALIMA in the event of a claim. Without an installation record, the conformity of the installation cannot be verified.
- Any incident, defect or abnormal behaviour must be reported to CB ALIMA without delay, and use of the function concerned suspended until it is resolved.
- These limitations do not apply in the event of personal injury, gross negligence or wilful misconduct, or where the law prohibits them.

2-year warranty

The ALIMA Simply board is guaranteed for 2 years against manufacturing and component defects, from the delivery date shown on the invoice.

Condition. The 2-year commercial warranty applies only if the installation record, complete and signed, is sent to contact@cb-alima.fr within 10 days of installation. The date the e-mail is received is authoritative. CB ALIMA acknowledges receipt. A record

sent late, incomplete or after a failure has occurred does not give entitlement to this commercial warranty; statutory warranties remain applicable.

What the warranty covers. Repair or replacement of the defective board, at CB ALIMA's option. A repaired or replaced board keeps the remaining warranty period.

How to claim. Write to contact@cb-alima.fr with the board serial number, the invoice and a description of the problem (photos of the installation if possible). The return of the board is agreed with CB ALIMA. Shipping of the board to CB ALIMA is paid by the customer; CB ALIMA pays for returning the repaired or replaced board if the warranty applies.

Exclusions. The warranty does not cover:

- an installation that does not comply with this manual: wrong connection, polarity or voltage outside specification, missing or unsuitable fuse;
- overloading of the outputs (more than 10 A per relay contact) or a short circuit;
- water, moisture, impacts, excessive heat, corrosion or surges on the vehicle's electrical system;
- any modification or repair by a third party, including adjusting the sealed regulator potentiometer;
- firmware not supplied by CB ALIMA;
- an accident, fire, transport damage or force majeure;
- normal wear of relays, terminals, connectors and cables;
- a board whose serial number is missing or illegible;
- removal and refitting costs, labour and vehicle downtime.

This commercial warranty is in addition to, and does not exclude, the statutory warranties.

Governing law. These conditions are governed by French law. Between professionals, any dispute falls within the jurisdiction of the competent courts for the registered office of CB ALIMA, after an attempt at amicable settlement.

This manual is an integral part of the product. It must be read before installation, kept by the installer and handed over with the vehicle. Only the current version, available from CB ALIMA, is binding; in case of any difference between this translation and the French version, the French version prevails. ALIMA Simply installer manual, version 2026-10.

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Installation record — ALIMA Simply

To be filled in and signed by the installer for each vehicle, then kept with the configuration file.
Send to contact@cb-alima.fr within 10 days of installation: condition of the 2-year warranty.

1. Identification

Installing company

Technician

Installation date

Registration number

Vehicle make and model

VIN

Board serial number

Firmware version

Name of the exported configuration file

2. Connections and fuses

Circuit	Connected function	Load (W)	Fuse (A)	Cable (mm²)
Power supply	Board power supply		2	
SORTIE 1	Left turn signal			
SORTIE 2	Right turn signal			
SORTIE 3				
SORTIE 4				
SORTIE 5				
SORTIE 6				

3. Checks before handing over the vehicle

- ☐ Wiring checked: no pinched or stripped cable, no contact with hot or moving parts
- ☐ Connections soldered or crimped and insulated, no self-stripping connectors
- ☐ All fuses fitted with the correct rating, as close as possible to the source
- ☐ Each ENTRÉE (1 to 4) triggers the correct function with its switch
- ☐ Each SORTIE controls the correct load and nothing else (application output test)
- ☐ ENTRÉE 12 V (feux): the function responds correctly with lights on, then off
- ☐ The vehicle's original control still works and keeps priority
- ☐ No warning light on the dashboard, no abnormal battery drain when parked
- ☐ Tests stationary with engine running, then road test, carried out by the installer
- ☐ Configuration exported and kept with this record
- ☐ Operation and limits explained to the customer, manual handed over

4. Remarks

5. Validation

Name and position of signatory

Date

Signature and company stamp

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